

Module 12 – DevOps and CI/CD Overview

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Introduction to DevOps

What is DevOps?

DevOps is a software development approach that combines **Development (Dev)** and **Operations (Ops)** to improve collaboration between software developers and IT operations teams. It enables organizations to develop, test, deploy, and maintain applications more quickly and reliably.

Instead of working separately, developers and operations teams work together throughout the software development lifecycle.

Goals of DevOps

The main goals of DevOps are:

- Faster software delivery
- Improved collaboration between teams
- Reduced deployment failures
- Continuous feedback
- High-quality software
- Faster recovery from failures
- Automation of repetitive tasks

Benefits of DevOps

- Faster application deployment
- Improved product quality
- Increased customer satisfaction
- Better communication among teams
- Reduced manual work through automation
- Continuous monitoring of applications
- More reliable software releases

Key DevOps Practices

1. Continuous Integration (CI)

Developers frequently merge code into a shared repository where automated builds and tests are executed.

2. Continuous Delivery (CD)

Code changes are automatically prepared for release so they can be deployed quickly.

3. Continuous Deployment

Every successful code change is automatically deployed to production without manual intervention.

4. Infrastructure as Code (IaC)

Infrastructure such as servers and networks is managed using code instead of manual configuration.

5. Continuous Monitoring

Applications and infrastructure are continuously monitored to detect issues early.

6. Automation

Testing, deployment, and infrastructure management are automated to reduce human errors.

Understanding CI/CD

What is Continuous Integration (CI)?

Continuous Integration is the practice of frequently integrating code changes into a shared repository. Every integration automatically triggers builds and tests.

CI Process

Developer → Git Repository → Automated Build → Automated Testing

Advantages of CI

- Early bug detection
- Faster development
- Better code quality
- Reduced integration issues
- Automated testing

What is Continuous Delivery (CD)?

Continuous Delivery ensures that software is always ready for deployment. Every successful build is automatically prepared for release.

Deployment to production still requires manual approval.

What is Continuous Deployment?

Continuous Deployment automatically deploys every successful build to production without human approval.

Difference Between Continuous Delivery and Continuous Deployment

Continuous Delivery	Continuous Deployment
Manual approval before deployment	No manual approval
Release when needed	Automatic release
Human decision required	Fully automated

Difference Between CI and CD

Continuous Integration	Continuous Delivery
Integrates code frequently	Delivers software automatically
Focuses on building and testing	Focuses on deployment
Detects bugs early	Ensures software is always release-ready

Benefits of CI/CD

- Faster software releases
- Improved software quality
- Automated testing
- Reduced deployment risks
- Faster feedback
- Better collaboration
- Reduced manual effort
- Increased customer satisfaction

Popular CI/CD Tools

1. Jenkins

- Open-source automation server
- Automates building, testing, and deployment
- Supports thousands of plugins

Features

- Open source
- Easy integration
- Large plugin ecosystem
- Highly customizable

2. GitHub Actions

GitHub Actions is GitHub's built-in CI/CD platform that automates workflows directly from GitHub repositories.

Features

- Easy workflow creation
- GitHub integration
- Supports Linux, Windows, and macOS
- Automatic testing and deployment

3. GitLab CI/CD

GitLab provides built-in CI/CD pipelines within GitLab repositories.

Features

- Integrated DevOps platform
- Automated pipelines
- Security scanning
- Continuous deployment

4. CircleCI

CircleCI is a cloud-based CI/CD platform that automates software development workflows.

Features

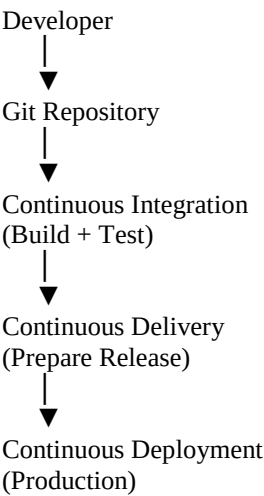
- Fast builds
- Cloud-based

- Docker support
- Easy integration with GitHub

Comparison of CI/CD Tools

Tool	Open Source	Cloud Support	Best Use
Jenkins	Yes	Yes	Enterprise automation
GitHub Actions	No	Yes	GitHub projects
GitLab CI/CD	Yes	Yes	GitLab repositories
CircleCI	No	Yes	Fast cloud CI/CD

CI/CD Workflow



Conclusion

DevOps combines development and operations to improve collaboration, automate workflows, and deliver software faster. CI/CD is an essential DevOps practice that automates building, testing, and deployment, helping organizations deliver reliable software efficiently. Popular tools like Jenkins, GitHub Actions, GitLab CI/CD, and CircleCI simplify the automation process and improve software quality.